

Silver Layer ETL: Documentation

Purpose of Silver Layer

The Silver Layer is a cleaned, standardized, and enriched version of the raw HR data from the Bronze Layer. The goal of this ETL (Extract, Transform, Load) process is to:

1. **Ensure data quality:** Remove inconsistencies, invalid values, and standardize formats.
 2. **Prepare data for reporting:** Make HR tables ready for analytics, dashboards, and business intelligence.
 3. **Maintain historical accuracy:** Preserve important HR attributes while correcting obvious errors.
 4. **Provide derived categories:** Add useful business categories (age group, tenure group, engagement level) for analysis.
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General Workflow

1. **Truncate existing Silver tables** – This ensures no duplicate or old data remains.
 2. **Extract data from Bronze tables** – Read raw HR tables (employee, recruitment, training, engagement).
 3. **Transform and clean data** – Correct values, standardize codes, replace missing data, and create new categories.
 4. **Load into Silver tables** – Insert the cleaned and transformed data into Silver tables.
 5. **Log ETL duration** – Keep track of how long each table took to load.
 6. **Error handling** – Catch errors during the ETL and print descriptive messages.
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Detailed Table-Level Operations

1. hr_employee_data

- **Source Table:** `bronze.hr_employee_data`
- **Target Table:** `silver.hr_employee_data`

Transformations & Cleaning:

1. Business Unit Standardization:

- Original short codes (e.g., NEL, CCDR) replaced with full descriptive names for readability.

2. Employee Status Logic:

- If the employee is active but has a termination type like Retirement or Resignation → mark as `Deactive`.
- If employee status = `Future Start` → mark as `Offer Cancelled`.

3. Termination Type Cleaning:

- Replace unknown termination (`Unk`) with `Not Yet`.

4. Termination Description:

- Replace NULLs with `N/A` for clarity.

5. State Standardization:

- Two-letter state codes converted into full state names.

6. Job Function Description:

- Abbreviations (CIO, CFO, VP) replaced with full titles.

7. Performance Score Standardization:

- If PerformanceScore = `PIP`, change to `Performance Improvement Plan`.

Key Outcome:

- Employee table is now consistent, readable, and ready for reporting on employee status, department, job, and performance.
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2. hr_employee_engagement_survey_data

- **Source Table:** bronze.hr_employee_engagement_survey_data
- **Target Table:** silver.hr_employee_engagement_survey_data

Transformations & Cleaning:

1. No major data cleaning required; this table mainly involves numeric survey scores.
2. Loaded the following fields:
 - EmployeeID
 - SurveyDate
 - EngagementScore
 - SatisfactionScore
 - WorkLifeBalanceScore

Key Outcome:

- Survey data is ready for analytics on employee engagement, satisfaction, and work-life balance.
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3. hr_employee_training

- **Source Table:** bronze.hr_employee_training
- **Target Table:** silver.hr_employee_training

No calculation has been done....

4. hr_recruitment_data

- **Source Table:** `bronze.hr_recruitment_data`
- **Target Table:** `silver.hr_recruitment_data`

Transformations & Cleaning:

1. Phone Number Cleaning:

- Invalid characters replaced or reformatted for consistency (e.g., `#` → N/A, `.` → `-`).

2. State Standardization:

- All state codes converted to full state names.

3. Job Title Cleanup:

- Remove unnecessary quotes from JobTitle.

4. Previous Job Department Extraction:

- Some data stored in the Status field is split into "Previous Job Department" and "Application Status."
- If values are standard statuses like `Interviewing`, `Applied` → mark department as `Not Mentioned`.

5. Application Status Logic:

- After extracting department info, the remaining Status field becomes the Application Status.

Key Outcome:

- Recruitment data is ready for analytics, showing applicants' previous departments, current job, and cleaned contact info.
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Additional Features Across All Tables

- **ETL Logging:**
 - Each table logs start and end time.
 - Duration for each table load is printed for monitoring.
 - **Error Handling:**
 - TRY...CATCH block ensures errors are captured with clear messages including:
 - Error message
 - Error number
 - Error state
 - **Consistency:**
 - All nulls, unknowns, and inconsistent codes replaced or standardized.
 - All names, states, job titles, and business units are human-readable.
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Business Benefits of Silver Layer

1. **Ready-to-use analytics tables** – Perfect for dashboards or reporting without extra cleaning.
2. **Consistent data across HR functions** – Employee, recruitment, training, and engagement data all standardized.
3. **Supports derived insights** – Age groups, tenure, training duration, engagement levels, and satisfaction levels are pre-calculated.

4. **Traceable ETL process** – Logs and error handling help maintain reliability and troubleshooting.

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